



Academic Year: 2024-25

Skill Enhancement Course-I

S.No.	Sem & Section	Title	Date (From -to)
1.	III SEM CSE-A,B,C&D	Python Programming	06/08/2024 -12/08/2024

Skill Enhancement Course-II

S.No.	Sem & Section	Title	Date (From -to)
1.	IV SEM CSE-A,B,C&D	Full Stack Development-I	20/12/2024 -12/04/2025

Skill Oriented Course-III

S.No.	Sem & Section	Title	Date (From -to)
1.	V SEM CSE-A,B,C&D	Cyber Security & Block chain	23/09/2024 - 26/09/2024, 28/09/2024 and 30/09/2024
		GenAI & Prompt Engineering	
		Django & Flask Framework	

Skill Oriented Course-IV

S.No.	Sem & Section	Title	Date (From -to)
1.	VI SEM CSE-A,B,C&D	React JS +Next JS + SQL	03/03/2025 -08/03/2025
		Spring Boot+MySQL	

Skill Oriented Course-V

S.No.	Sem & Section	Title	Date (From -to)
1.	VII SEM CSE-A,B,C&D	Aptitude & Verbal	06/08/2024 - 10/08/2024

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Python Programming

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate basic concepts and control structures in python Programming. **(K3)**

CO2: Demonstrate functions and packages. **(K3)**

CO3: Construct python programs using structured data types. **(K3)**

CO4: Develop programs on Files, Exception handling and OOPs Concepts. **(K3)**

CO5: Construct programs for Data Analysis using Num Py and Pandas. **(K3)**

Course Contents:

- Basics of Python Programming
- Control Flow Statements
- Functions, Strings and Lists
- Python Data Structures : Dictionaries, Tuples , and Sets
- Files and Object-Oriented Programming
- Introduction to Data Science



Full Stack Development-I

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate HTML elements and their attributes for designing static web pages. **(K3)**

CO2: Apply appropriate CSS styles to HTML elements. **(K3)**

CO3: Demonstrate JavaScript Pre-defined and User-Defined Objects. **(K3)**

CO4: Develop dynamic web pages and validate forms using JavaScript. **(K3)**

Experiments covering the Topics:

- Lists, Links and Images
- HTML Tables, Forms and Frames
- HTML 5 and Cascading Style Sheets, Types of CSS
- Selector forms
- CSS with Color, Background, Font, Text and CSS Box Model
- Applying JavaScript - internal and external, I/O, Type Conversion
- JavaScript Conditional Statements and Loops, Pre-defined and User-defined Objects
- JavaScript Functions and Events
- Node.js



Cyber Security & Block chain

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Understand core cyber security concepts and identify various cyber threats.. **(K2)**

CO2: Implement network and application security measures to mitigate vulnerabilities. **(K3)**

CO3: Discuss cryptographic principles and their role in blockchain technology. **(K3)**

CO4: Develop and deploy smart contracts and decentralized applications on blockchain platforms. **(K3)**

CO5: Analyze and apply security auditing and risk management techniques to ensure compliance with industry standards. **(K3)**

Course Contents:

- Introduction to Cyber Security
- Network Security & Wireless Networks
- Secure Software Development & Web Application Security
- Introduction to Cryptography
- Cyber Security Practices
- Hands-on Session Cyber Security
- Introduction to Blockchain
- Cryptography and Consensus Algorithms
- Smart Contracts and Decentralized Applications
- Hands-on Session Smart Contracts and Decentralized Applications
- Blockchain Platforms and Frameworks
- Use cases on Blockchain

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Gen AI & Prompt Engineering

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Explain the fundamentals of Generative AI and Machine Learning. **(K2)**

CO2: Apply classification algorithms and evaluate their performance. **(K3)**

CO3: Design and implement neural networks for deep learning tasks. **(K3)**

CO4: Produce images from text prompts using prebuilt text-to-image models. **(K3)**

CO5: Develop a basic chatbot using large language models. **(K3)**

Course Contents:

- Introduction to Gen AI
- Introduction to ML
- Performance Metrics in ML
- ML with Hands-on
- Deep Learning Overview
- ANN with Hands-on
- CNN with Hands-on
- RNN with Hands-on
- NLP Fundamentals
- NLP Hands-on
- GANs with Hands-on
- AEs with Hands-on
- Exploration of Hugging Face
- Pre-built Text-to-Image Models
- Hands-on Activity
- LLMs
- Prompt Engineering
- Custom Chatbot using Hugging Face Text-to-Text Models
- Custom Chatbot using APIs

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Django & Flask Framework

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Explain Django and Flask project structure and workflow. **(K2)**

CO2: Demonstrate Django & Flask Templates. **(K3)**

CO3: Illustrate Django ORM & SQLAlchemy ORM. **(K3)**

CO4: Develop User Authentication and Authorization in Django and Flask. **(K3)**

CO5: Demonstrate Django REST Framework and Flask Extensions. **(K3)**

Course Contents:

- Introduction to Flask Framework
- Introduction to Django Framework
- Hands-on with Django Framework
- Hands-on with Flask Framework
- Django Models and Database Integration
- Working with Databases in Flask
- Performing CRUD Operations
- Working with Django Forms
- User Authentication and Authorization
- User Authentication in Flask
- Flask Extensions and API Development
- Django REST Framework and Deployment



React JS +Next JS + SQL

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Explain the fundamental concepts of React JS, Next JS, and SQL including component structure, routing mechanisms, and database operations. **(K2)**
- CO2:** Demonstrate the use of React hooks, Next.js data fetching methods, and SQL queries for data retrieval and manipulation.. **(K3)**
- CO3:** Apply state management, routing, and component-based design to develop responsive user interfaces using React JS and Next JS. **(K3)**
- CO4:** Develop a full-stack application integrating React/Next.js frontend with a SQL-based backend, using appropriate API endpoints.. **(K3)**
- CO5:** Apply client-side and server-side data fetching techniques using Fetch API, SWR, and Next.js methods to interact with backend services and databases.. **(K3)**

React JS:

- **Introduction to React & Setup**
 - What is React? Why use it?
 - Setting up a development environment (Node.js, npm/yarn, VS Code).
 - Creating a basic React application (create-react-app).
 - Understanding JSX.
- **Components & Props**
 - Functional vs. Class components.
 - Props: Passing data between components.
 - Component composition.
 - Basic Styling.
- **State & Lifecycle**
 - Managing component state using `useState`.
 - Understanding component lifecycle methods (`useEffect`).
 - Event handling.
- **Lists & Keys, Conditional Rendering**
 - Rendering lists of data.
 - Using keys for efficient rendering.
 - Conditional rendering based on state.
 - Basic forms.
- **Hooks & Context API**
 - Deep dive into common hooks (`useState`, `useEffect`, `useContext`).
 - Using the Context API for state management.
 - Custom hooks.
- **Routing (React Router)**
 - Setting up React Router.
 - Creating routes and navigation.
 - Passing parameters in routes.

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Next JS

- **Introduction to Next.js & Setup**
 - What is Next.js? Why use it?
 - Setting up a Next.js project.
 - Understanding pages and routing.
 - File-based routing.
- **Data Fetching (getServerSideProps, getStaticProps)**
 - Server-side rendering (SSR) and static site generation (SSG).
 - Fetching data using `getServerSideProps` and `getStaticProps`.
 - Understanding when to use each approach.
- **API Routes**
 - Creating API routes in Next.js.
 - Handling HTTP requests.
 - Integrating with backend services.
- **Client-Side Data Fetching (SWR or Fetch)**
 - client side data fetching inside of nextjs.
 - using SWR or standard fetch.
- **Styling in Next.js**
 - CSS modules.
 - Styled components.
 - Tailwind CSS.
- **Deployment**
 - Deploying Next.js applications to platforms like Vercel or Netlify.

SQL

- **Introduction to SQL & Database Setup**
 - Relational databases and SQL basics.
 - Setting up a local SQL database (SQLite or PostgreSQL).
 - Creating tables and defining data types.
- **Basic SQL Queries (SELECT, WHERE, ORDER BY)**
 - Retrieving data using SELECT statements.
 - Filtering data using WHERE clauses.
 - Sorting data using ORDER BY.
- **Data Manipulation (INSERT, UPDATE, DELETE)**
 - Inserting new data into tables.
 - Updating existing data.
 - Deleting data.
- **Joins & Relationships**
 - Understanding table relationships (one-to-one, one-to-many, many-to-many).
 - Joining tables using INNER JOIN, LEFT JOIN, etc.

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Department of Computer Science & Engineering (Accredited by NBA)

- **Connecting React/Next.js to SQL**
 - Using Node.js as a backend.
 - Connecting Node.js to a SQL database.
 - Creating API endpoints to interact with the database.
- **Project: Building a Full-Stack Application**
 - Putting it all together: building a simple application that uses React/Next.js for the frontend and SQL for the backend.

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Spring Boot+MySQL

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Explain the fundamental concepts of Spring Boot, dependency injection, and the architecture of RESTful APIs. **(K2)**
- CO2:** Describe how Spring Boot integrates with MySQL using Spring Data JPA for data persistence and database operations.. **(K2)**
- CO3:** Develop RESTful web services using Spring Boot, handling HTTP requests and performing CRUD operations with MySQL.. **(K3)**
- CO4:** Apply advanced features like entity relationships, custom queries, validation, and exception handling in a Spring Boot application.**K3)**
- CO5:** Apply a secure, tested, and documented RESTful API application by integrating Spring Security, Swagger, and testing frameworks like JUnit and Mockito. **(K3)**

- **Introduction to Spring Boot & Setup**
 - What is Spring Boot? Why use it?
 - Setting up a development environment (Java, Maven/Gradle, IntelliJ IDEA/Eclipse).
 - Creating a basic Spring Boot application.
- **Spring Boot Basics (Dependency Injection, Annotations)**
 - Understanding dependency injection.
 - Using Spring Boot annotations (@Component, @Service, @Controller, etc.).
 - Creating beans.
- **RESTful APIs (Controllers, Request/Response Handling)**
 - Building RESTful APIs using Spring MVC.
 - Handling HTTP requests (GET, POST, PUT, DELETE).
 - Serializing and deserializing JSON.
- **Data Persistence with Spring Data JPA**
 - Introduction to JPA and Hibernate.
 - Setting up Spring Data JPA.
 - Defining entities and repositories.
- **MySQL Database Setup & Integration**
 - Setting up a local MySQL database.
 - Connecting Spring Boot to MySQL.
 - Configuring database properties.
- **CRUD Operations with Spring Data JPA & MySQL**
 - Implementing CRUD operations (Create, Read, Update, Delete) using Spring Data JPA.
 - Testing API endpoints.
- **Advanced JPA (Relationships, Queries)**
 - Managing entity relationships (one-to-one, one-to-many, many-to-many).
 - Writing custom queries using JPQL or native SQL.
- **Exception Handling & Validation**
 - Handling exceptions in RESTful APIs.
 - Validating request data.
 - Global exception handling.

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- **Security (Spring Security)**
 - Introduction to Spring Security.
 - Implementing basic authentication and authorization.
 - JWT Authentication.
- **Testing (JUnit, Mockito)**
 - Writing unit tests for Spring Boot applications.
 - Using Mockito for mocking dependencies.
 - Integration testing.
- **Logging & Monitoring**
 - Implementing logging in Spring Boot.
 - Basic monitoring.
- **API Documentation (Swagger/OpenAPI)**
 - Generating API documentation using Swagger/OpenAPI.
- **Project: Building a RESTful API Application**
 - Combining all skills learned into a complete project.

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Aptitude & Verbal

Course Outcomes: After successful completion of the Course, the student will be able to:

- Use their Logical thinking and analytical abilities to solve quantitative aptitude questions form Company specific and other competitive tests. [K3]
- Solve questions relative to Time & distance and time & work etc. from company specific. [K2]
- Solve puzzle related questions for any type of competitive tests. [K2]
- Practice the Aptitude Round Clearing ability in interview Process. [K3]
- Analyze discourse markers to speak clearly on a specific topic in informal discussions. [K3]

Course Contents:

Aptitude:

- Number System
- Percentages
- Ration and proposition
- Averages
- Time and work
- Time and Distance
- Profit and loss
- Simple Interest and Compound Interest
- Permutations and combination
- Probability
- Data Analysis
- Series
- Analogy
- Directions and classifications
- Blood Relations
- Syllogism
- Seating
- Circular
- comparisons
- Coding and Decoding
- Clock and calendar

Verbal:

- Grammar, Tenses
- Preposition & Articles
- Speech & Voices
- Reading Comprehension
- Sentence Correction and Sentence Selection
- Sentence Arrangement
- Spotting Error
- Synonyms & Antonyms

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